Progress in Additive Manufacturing (2025) 10:7947–7962
https://doi.org/10.1007/s40964-025-01084-7

RESEARCH

Experimental and numerical thermal monitoring of SS308L structures processed by wire arc additive manufacturing

Y. Osama[1] · M. Abdelwahed[1] · A. Elsabbagh[1] · M. A. Taha[1,2]

Received: 8 January 2025 / Accepted: 18 March 2025 / Published online: 15 April 2025
© The Author(s) 2025

Abstract

Wire arc additive manufacturing (WAAM) is highly sought after for producing complex-shaped products with high deposition rates and relatively low cost compared to other additive manufacturing processes. However, the cumulative heat input during processing leads to low-dimensional accuracy, high distortion, and high residual stress, which requires heat control and thermal monitoring. The current work offers a study on the applicability of experimental and numerical thermal monitoring in detecting thermal effects generated during the WAAM processing of a corrosion-resistant SS308L austenitic alloy commonly used in nuclear, aerospace, and marine industries. Temperatures were measured during processing for vertical and horizontal structures by a mono-wavelength pyrometer and coupled with thermo-mechanical simulations. The simulation models were successfully validated through the distortion measured in the substrates of the processed structures, confirming the acquired thermal cycles. Results showed that mono-wavelength pyrometry was only effective for vertical structures away from the deposition area, with emissivity varying from 0.6 to 0.41 for temperatures between 600 and 1700 °C, respectively. For horizontal structures, temperature measurements relied solely on simulations, which predicted heat accumulation and cooling rates. These predictions were validated through secondary dendritic arm spacing and hardness measurements, confirming the impact of intermittent deposition in reducing heat build-up and leading to relatively finer microstructures and less variant hardness profile compared to continuous deposition.

✉ Y. Osama
yomna.osama@eng.asu.edu.eg

M. Abdelwahed
marwan.faisal@eng.asu.edu.eg

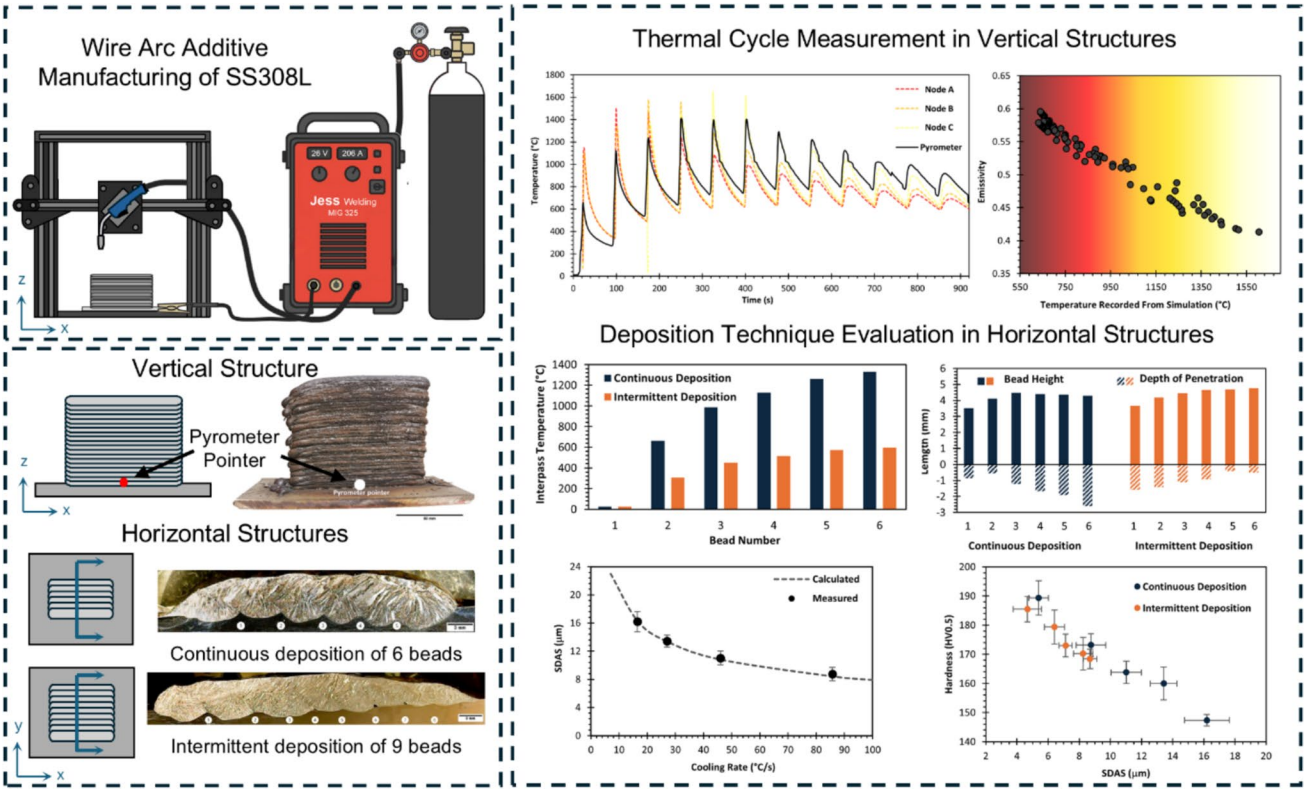
A. Elsabbagh
elsabbagh.ahmed@eng.asu.edu.eg

M. A. Taha
mohamed_taha@eng.asu.edu.eg

[1] Ain Shams University, Cairo, Egypt

[2] Academy of Scientific Research and Technology (ASRT), Cairo, Egypt

Graphical abstract



Keywords Wire arc additive manufacturing · Stainless steel · Microstructure · Finite element simulation · Thermal monitoring

1 Introduction

With the increasing demand for additive manufacturing, wire arc additive manufacturing (WAAM) is playing a vital role in fulfilling the current needs for rapid prototyping/manufacturing of products required in various industrial sectors. Owing to its relatively low investment cost and high deposition rate, WAAM is preferred over other additive techniques to fabricate large-sized complex products [1]. In addition, various metals can be processed using WAAM, such as titanium alloys [2], aluminum alloys [3], nickel alloys [4], steels [5], and stainless steels [6].

Stainless steels are used extensively in modern production due to their excellent mechanical properties and corrosion resistance, providing long-term sustainability in extreme environments [7]. Altering the chemical composition of the alloying elements stabilizes different phases, resulting in various grades, such as austenitic, ferritic, martensitic, duplex, and precipitation hardening, each designed for particular use [8]. Austenitic stainless steels, including 304L, 308L, and 316L, are mostly compatible with additive manufacturing, especially WAAM, as they exhibit good weldability and are widely accessible in suitable feedstock form [9]. Furthermore, the distinctive combination of corrosion resistance, mechanical strength, and low-temperature properties makes them perfect candidates for products in the nuclear [10], aerospace, automotive, marine, and chemical industries [11].

Generally, the applications for WAAM benefit several industries, such as bulbous bows, propellers, and rudders for the shipbuilding industry [12], aircraft spars, fuel tanks, airframes, and satellite breadboards for the aerospace industry [13], gusset plates and stiffening elements for the automotive industry [14], columns, beams, frames, gusset plates, trusses, complex joints and nodes for the construction industry [15], in addition to mixing tanks and transport vessels for pharmaceutical and food industry [16].

Despite the wide range of applications for WAAM, this technology still suffers from several drawbacks related to low-dimensional accuracy and surface quality due to high surface roughness and irregularities during the deposition

process [17], defects, such as cracks and porosity [18], and residual stresses and strains due to the heat input during metal deposition [19]. For instance, high heat input results in a large bead width and height, which in turn increases the surface roughness of the final product [20]. Additionally, mechanical properties, such as hardness and tensile strength, decrease with increased heat input due to grain coarsening [21]. On the other hand, lower heat inputs can be insufficient to fully melt the deposited metal during bead overlap, which causes lack of fusion [22]. Therefore, effective thermal management is essential in maintaining the quality and reducing defects and distortion during WAAM. That can be achieved by adjusting the processing parameters or using heat source variants of Gas Metal Arc Welding (GMAW), Cold Metal Transfer (CMT), which was specially developed for WAAM to control heat input by switching between positive and negative polarity [23], or by introducing interpass active cooling methods [24]. Hence, the application of thermal monitoring becomes even more demanding for a more efficient WAAM process.

Thermal monitoring for WAAM process has taken two main approaches, which are in situ thermal monitoring using thermocouples, infrared cameras, and infrared pyrometers [25]; and computational methods using Finite Element Method (FEM) [26] and Artificial Neural Networks (ANN) [27]. The selection process of the in situ thermal monitoring technique depends on the advantages and disadvantages of each method. Thermocouples are relatively inexpensive, easy to implement, and provide highly accurate readings at a single point. However, positioning thermocouples are limited to substrate measurements as they may interfere with the WAAM process and delay reading acquisition, which is unstable for the highly dynamic metal deposition process [28]. On the other hand, infrared cameras and pyrometers are favored due to their real-time and contactless measurements, yet they suffer from multiple limitations in thermal monitoring of WAAM process. Their main drawback is the emissivity dependency, which is a function of the surface finish and the color of the metal that changes at different temperatures [29]. In addition, factors, such as oxidation and contamination, also influence emissivity and introduce measurement errors [30]. Lastly, the pyrometer cannot detect infrared radiation from the melt pool due to the intense light and electromagnetic radiation resulting from the welding arc [31]. Müller et al. [32] overcame the emissivity-temperature dependency by calibrating the emissivity of an infrared thermal camera using a dual-wavelength pyrometer measuring thermal cycles for a high-strength steel vertical structure during WAAM. It was found that the emissivity as a function of temperature stabilizes after the third layer, and a consistent function for layer three was used for corrections where the emissivity decreased with increasing temperature.

Computational methods can be superior to in situ thermal monitoring as they reduce time and costs by reducing the number of experimental trials, and they utilize thermal data to predict defects, such as residual stress and strains [26]. Several studies have used the thermal history recorded by simulation modeling for parametric studies or microstructure and mechanical properties correlations. For instance, recent simulation modeling studies have enlightened the effects of dwell time on the attributes of WAAM vertical structures. Li and Xiong [34] concluded that longer dwell times result in uniform stress distribution, reduce axial and longitudinal residual stresses, and aid in mitigating thermal cycling-induced distortions, improving the dimensional accuracy for thin vertical WAAM structures. Furthermore, Yang et al. [35] found that the application of dwell times minimizes re-melting and reduces thermal cycling, which leads to a further decrease in residual stress formation. Similarly, Abusalma et al. [36] studied the impact of dwell time on the residual stress of Inconel 718 deposited on A36 steel and discovered that shorter dwell times resulted in more consistent residual stress profiles. Additionally, with the aid of simulation modeling, Kumar et al. [37] examined the effect of dwell time (60, 90, 120, 150, and 180 s) for the SS316L vertical structure and found that an ideal dwell time of 120 s is sufficient to balance grain refinement and lowers heat build-up, improving mechanical properties. For SS308L, Le et al. [38] studied the effect of 30-s and 60-s dwell time compared to active air cooling on vertical structures. The main findings showed that active cooling slightly improves surface finish and mechanical properties compared to free cooling, though significantly enhances productivity as active cooling requires only 10 s per layer instead of longer dwell times.

Despite the extensive study on the WAAM vertical structures, limited information regarding the processing of horizontal structures by WAAM is found in the open literature to the authors' best knowledge. Additionally, in situ thermal monitoring for vertical structures using mono-wavelength pyrometers represents a significant challenge due to the emissivity dependence on the temperature during WAAM. Hence, this work aims to utilize thermo-mechanical simulation models as a tool for thermal monitoring during WAAM of SS308L vertical and horizontal structures. By correlating simulation results with in situ mono-wavelength pyrometer readings for vertical structures and microstructural analysis for horizontal structures, this study enhances the thermal assessment during WAAM while addressing the limitations of mono-wavelength pyrometry, particularly its dependence on temperature-dependent emissivity. In addition, the effects of dwell time of horizontal structure on the layer quality and mechanical properties were also studied. The selection of 308L grade for this study was based on higher ferrite content

compared to 304L, which prevents solidification cracking [39] and lower cost compared to 316L, to be used for applications where extreme corrosion resistance is not required, such as nuclear, aerospace, and marine industries.

2 Materials and methods

2.1 Feedstock material

A commercial stainless steel wire of grade SS308L having a diameter of 1.2 mm was used as a feedstock in the current study, in which the depositions were carried on a 6 mm-thick SS316L substrate. The chemical compositions of the deposited SS308L structures and the SS316L substrate were analyzed using SPECTRO – MAXX Spectrometer Arc/Spark OES, and collected in Table 1.

2.2 WAAM processing conditions

The WAAM setup used in the current investigation is composed of a MIG 325 Jess welding machine integrated with a Cartesian robotic system. Preliminary optimization was performed on single tracks by varying torch travel speed, wire feed speed, voltage, and standoff distance. The optimum parameters were selected based on a statistical analysis of the bead geometry consistency and porosity formation. Consequently, the WAAM processing parameters for fabricating vertical and horizontal structures were adjusted as follows: torch travel speed of 400 mm/min, wire feed speed of 9 m/min, standoff distance of 7 mm, and 99.999% purity argon as the shielding gas with a flow rate of 20 L/min. The average voltage and the current were 22 V and 206 A, respectively. Initial single bead experiments were performed using the given processing parameters, and the bead dimensions were found to be 8.5 mm in width, 3.2 mm in height, and 1.8 mm in depth, as presented in Fig. 1a.

Three identical vertical structures with a length of 100 mm and a height of 80 mm were processed using a bi-directional strategy, in which a dwell time of 60 s was applied for the subsequent layers. Additionally, two horizontal structures were deposited with a 40% overlap using a zigzag filling strategy, in which one structure was processed continuously while an intermittent deposition with a dwell time of 60 s was applied for the other. It is noteworthy to mention that the former condition enabled the processing of 6 beads (60 mm × 33.1 mm) only due to the massive heat accumulation in the substrate material, while a number of 9 beads (60 mm × 40.8 mm) were possible to be processed by the application of dwell time between the overlaps. In vertical structures, the thermal cycles during the WAAM processing were monitored using a mono-wavelength Extech 42,570-NISTL Dual Laser IR pyrometer having a spot size of 6 mm at the focal point, and the emissivity was set at a value of 0.44. The location of the thermal measurements is schematically presented in Fig. 1.

Table 1 Chemical composition of investigated SS308L feedstock and deposited structure

Element	C	Mn	Si	Cr	Ni	P	S	Mo
SS308L Wire Feedstock	≤0.03	1–2.5	0.3–0.65	19.5–22	9–11	≤0.03	≤0.03	–
As-deposited SS308L	0.0134	1.730	0.351	19.056	9.396	0.027	0.006	–
SS316L Substrate	0.0145	1.357	0.485	16.891	10.222	0.029	0.001	2.036

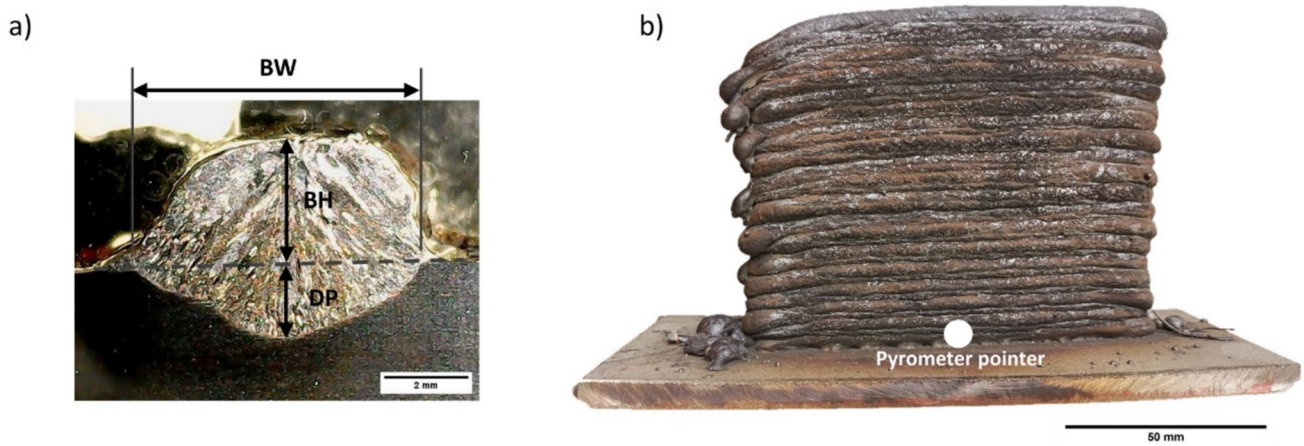


Fig. 1 Experimental work: **a** SS308L single bead dimension measurements and **b** Pyrometer position during thermal cycle measurements of SS308L vertical structure

2.3 Material characterization and mechanical testing

The WAAM processed specimens were prepared according to the standard metallographic procedures and followed by fine polishing with 50 nm-sized alumina suspension. In order to highlight the microstructural features, electro-etching was applied to the prepared specimens by immersing them in a nitric acid solution at a concentration of 67% and applying a direct current at 20 V. Images of the microstructures were acquired using an OPTIKA microscope equipped with OPTIKAM camera. Hardness tests were conducted using the Digital Vickers Hardness Tester TH721/721Z at a load of 0.5 kg for 15 s.

2.4 Thermal simulation modeling

Simufact Welding v2024.1 software was used to assess thermal cycles at various locations, using the heat source distribution parameters and the actual geometry of the experimental samples as initial inputs. The meshing of the deposited geometry was set to half the size of a single bead height, as illustrated in Fig. 2. The studied nodes were taken at the bottom of the horizontal structure, at a height equal to the bead height of a single bead, and at a height equal to double the bead height of a single bead. The models were validated by comparing the deformation in the substrate after metal deposition calculated by the FEM model, as shown in Fig. 3, with the measured experimentally, as shown in Table 2.

For the SS308L vertical structure, the thermal cycles from experimental and simulation results were compared to determine the relationship between recorded temperature and emissivity of SS308L stainless steel during WAAM using the following equations [40]:

$$\frac{1}{T_M^*} = \frac{1}{T_A} + \frac{\lambda_{\text{eff}}}{c_2} ln\varepsilon_k \tag{1}$$

$$T_M = T_M^* + \delta T \tag{2}$$

$$\delta T = \frac{\varepsilon_k - \varepsilon}{\varepsilon_k} * \frac{\lambda_{eff}}{c_2} * T_M^{*\,2} \tag{3}$$

where T_M is the temperature measured by the simulation model, T_M^* is a preliminary temperature, T_A is the raw temperature measured by the pyrometer, c_2 is the Planck's second radiation constant (1.4387×10^{-2} m.K), ε_k is the set emissivity value (0.44), ε is the actual emissivity, and λ_{eff} is the effective wavelength (10 µm).

3 Results and discussion

3.1 Thermal cycle of WAAM vertical structures

Three identical SS308L vertical structures were processed by WAAM, and the thermal cycles of the initial beads were recorded using a mono-wavelength pyrometer, Fig. 4. It could readily be observed that the thermal profiles of the processed structures are fairly consistent, which highlights the repeatability of the measurements. For the initial layers, it was found that incomplete temperature measurements were recorded since the investigated SS308L is expected to melt at around 1450 °C. This can be attributed to the arc generation and deposition dynamics itself, besides the relatively larger spot size of the pyrometer compared to the actual height of the deposited beads as schematically visualized in Fig. 5. Additionally, sooty surfaces formed during DED-Arc processes due to smoke and fumes can significantly affect the material's emissivity [41]. These deposits

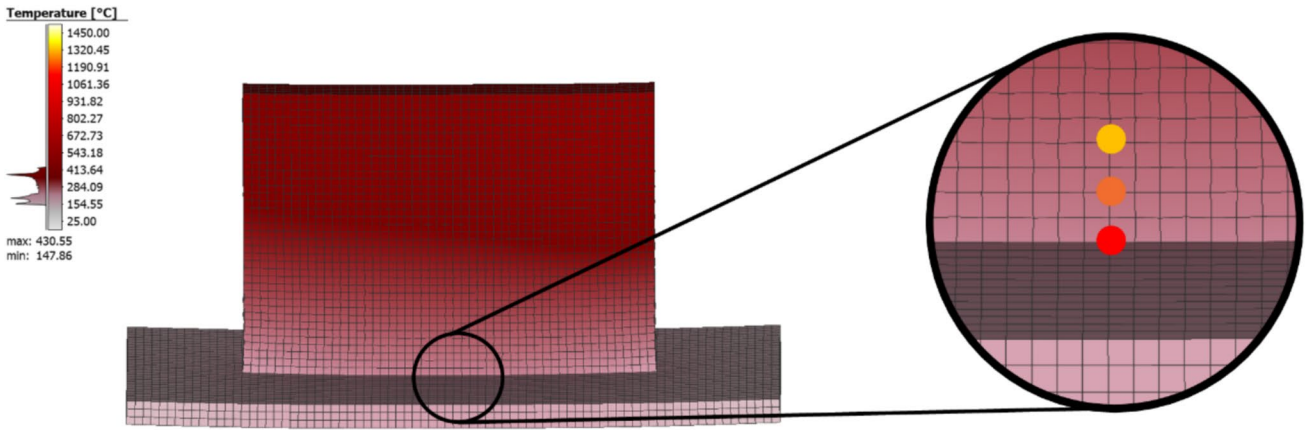


Fig. 2 WAAM FEM model and studied nodes for SS308L vertical structure

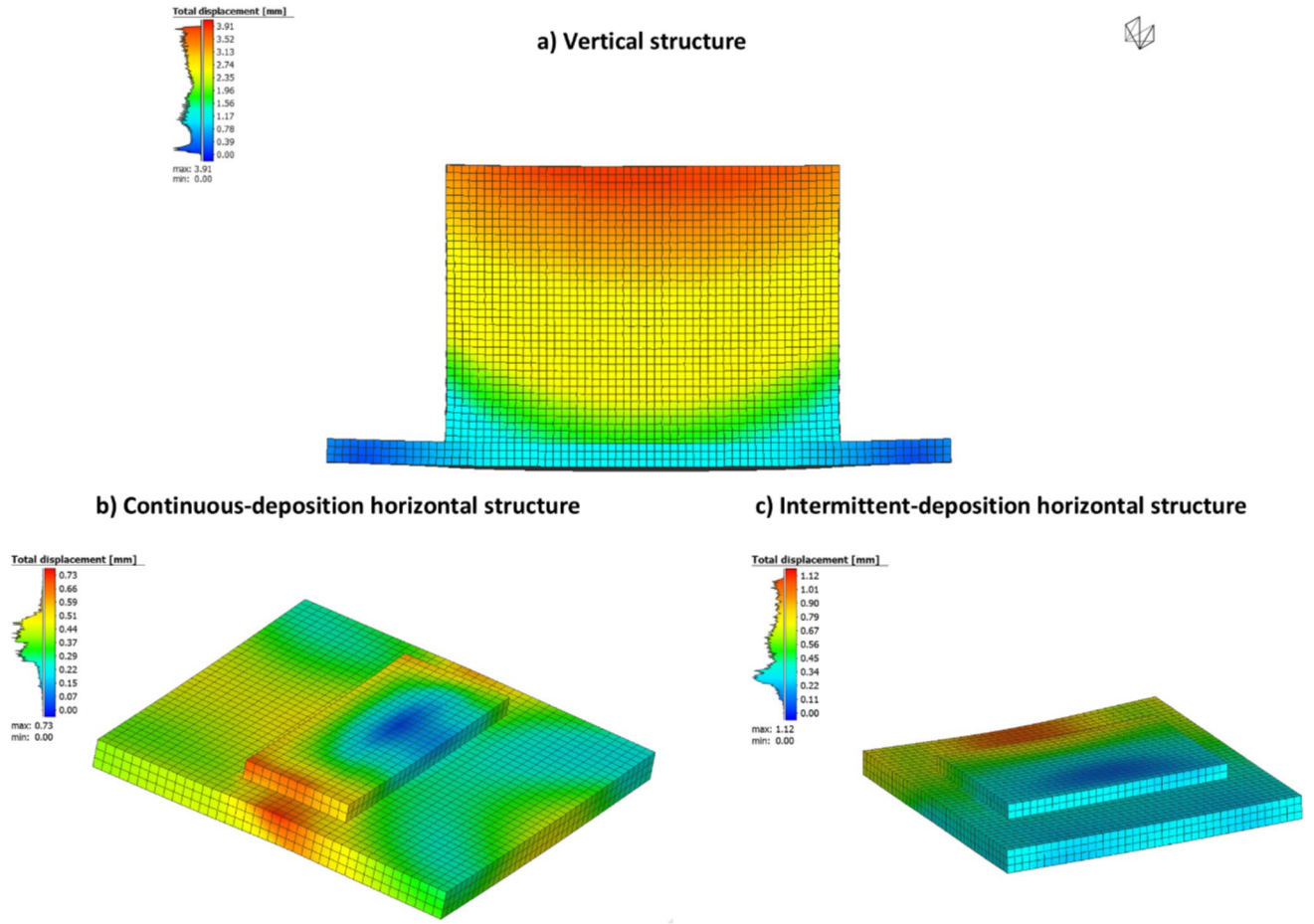


Fig. 3 Residual deformation from FEM model in SS308L **a** vertical structure, **b** continuous deposition horizontal structure, and **c** intermittent deposition horizontal structure

Table 2 The deflection values in the substrate: measured experimentally and calculated by the simulation model

Structure	Vertical	Continuous Deposition Horizontal		Intermittent Deposition Horizontal	
Substrate Deflection	Max. Deflection	Min. Deflection	Max. Deflection	Min. Deflection	Max. Deflection
Experimental (mm)	1.33	0.39	0.48	0.42	1.29
Simulation (mm)	1.37	0.37	0.44	0.34	1.12

and potential blockage of infrared radiation lead to inaccurate temperature readings in thermography [32]. From the cooling curves, the pyrometer measurements showed a thermal arrest-like (marked with arrows in Fig. 4) at ~ 1000 °C. This phenomenon was also reported by Felice et al.[42] for mono-wavelength pyrometer measurement when the emissivity is set at an intermediate value between the emissivity of solid material at different temperatures.

To get a reliable estimation of the temperature measurements during the WAAM processing of SS308L vertical structures, a FEM model was constructed, and the thermal history was simulated at three different surface nodes covering the spot size of the utilized pyrometer, Fig. 5. The experimental and numerical thermal profiles of the initial beads during the processing of 12 layers were collected for a vertical structure, as shown in Fig. 6. It could be observed that the numerical results provided temperature profiles similar to the experimental results. It is to be mentioned that during the 1st heating cycle, the simulated temperature does not reach the expected melting temperature as the imported geometry (Fig. 7) vertical structure was assumed to have a constant bead width along the built layers for simplicity purposes. Moreover, it was found that the deposited beads undergo roughly 2 ~ 3 times re-melting due to the heat

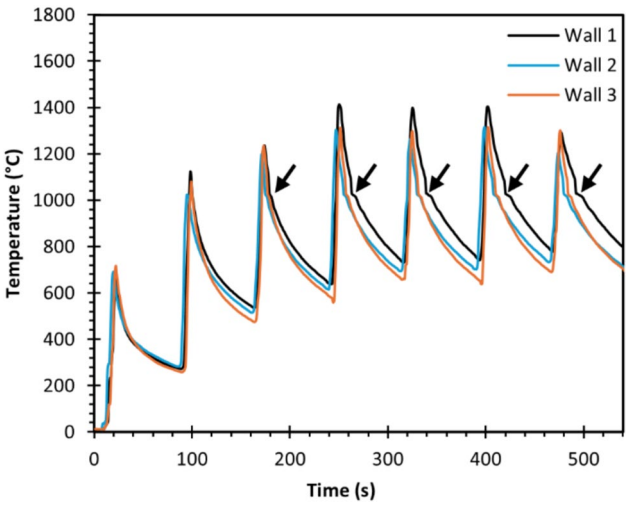


Fig. 4 The recorded thermal cycles of the three SS308L vertical structures using a mono-wavelength pyrometer

accumulation and the successive deposition of layers [39], after which the beads could reach a steady plateau at a temperature around 600 °C.

The pyrometer readings show thermal patterns comparable to the simulation results, especially at Node C, starting from the 4th thermal cycle. Accordingly, thermal cycles obtained experimentally and numerically at Node C were substituted in Eqs. (1–3) to calculate the corrected emissivity. Figure 8 illustrates the relationship between the emissivity of SS308L stainless steel during the deposition process and the temperature recorded by the FEM model. The results show a linear decrease in emissivity value as the temperature increases, ranging from 0.6 at low temperatures to 0.41 near the melting temperature. Once the melting temperature is reached, the emissivity value remains constant. This observed change in emissivity is attributed to the changes in the metal surface during heating, particularly the development of annealing colors at temperatures above 550 °C. Thus, the selection of the emissivity value during in-process thermal monitoring depends on the desired range of temperatures to be accurately measured. For SS308L, Fig. 8 can be used as a selection guide for the emissivity of a mono-wavelength pyrometer to thermally monitor structures at specific temperatures during WAAM process.

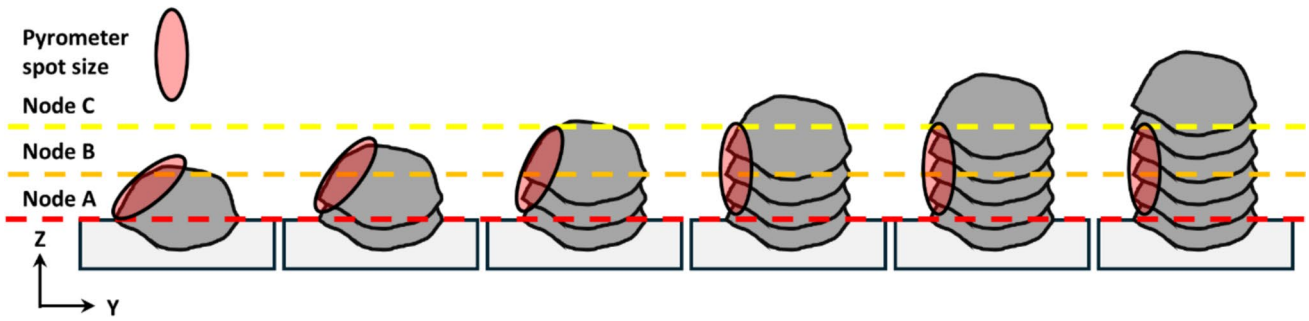
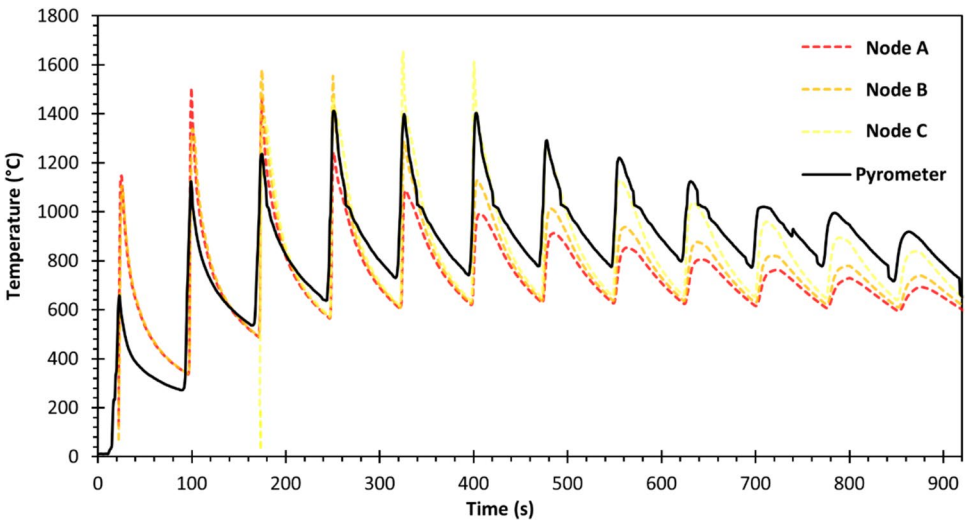


Fig. 5 Visualization of thermal cycle acquisition experimentally and numerically during SS308L vertical structure deposition

Fig. 6 Thermal cycles recorded by mono-wavelength pyrometer at the first layers and FEM model at three different nodes



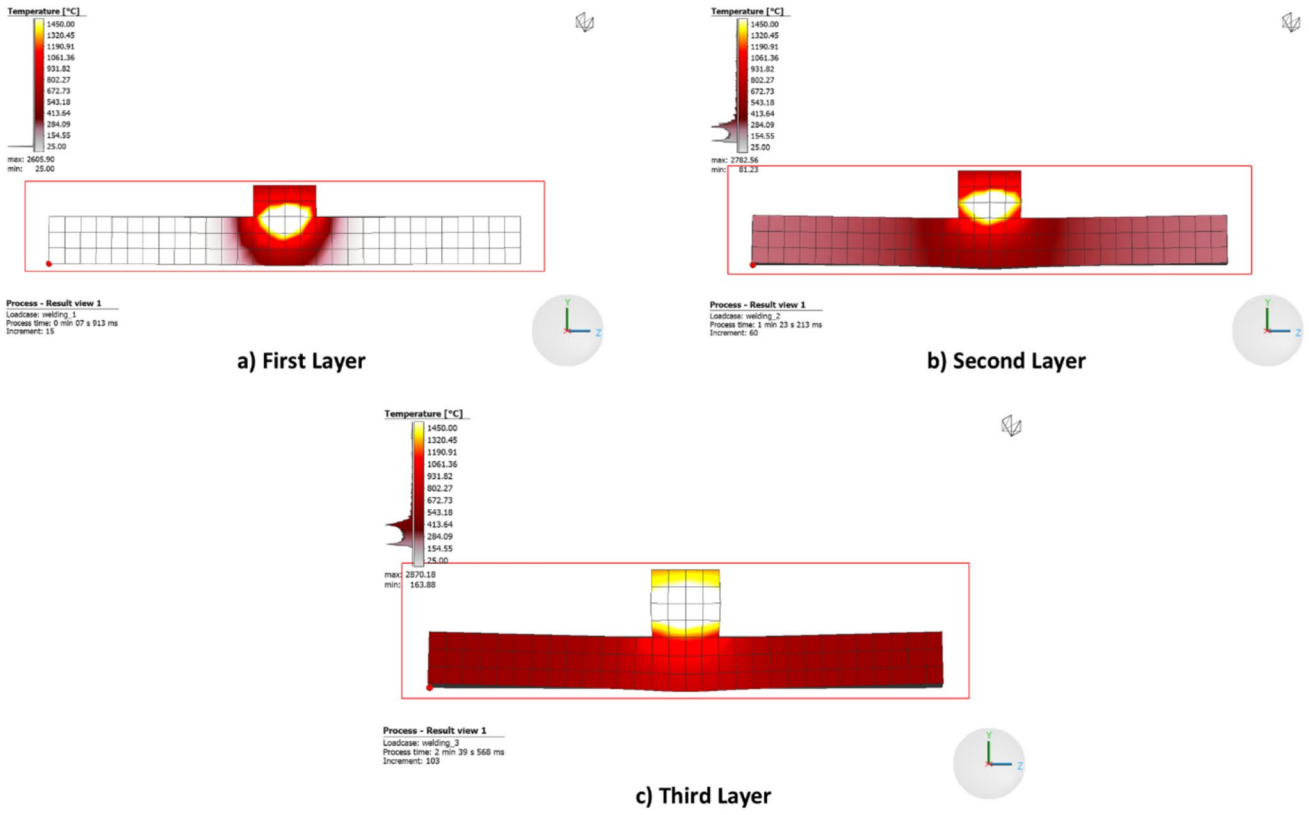


Fig. 7 Heat source dimensions during the deposition of **a** the first layer, **b** the second layer, and **c** the third layer of the SS308L vertical structure

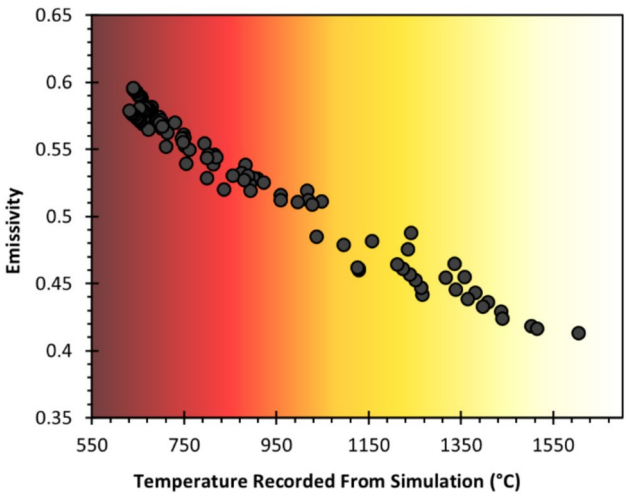


Fig. 8 The relation between the emissivity of the SS308L vertical structure during WAAM and the temperatures obtained from the FEM model

3.2 Thermal cycle of WAAM horizontal structures and effects of dwell time

The main findings of the above-mentioned study showed that the results of the constructed FEM model are comparable to the experimental measurements and could provide more reliable results, especially in the fusion zone. Hence, horizontal structures composed of overlapped beads were processed by WAAM, and the thermal history of the processed beads was simulated at the interpass of the beads, as shown in Fig. 9. It is to recall that a number of six beads were only possible to be processed by a continuous deposition, after which the substrate material starts to melt due to the massive heat accumulation. Therefore, a dwell time of 60 s was also considered in the experiments to potentially reduce the heat accumulation in the substrate and allow the processing of larger structures.

Thermal cycles at overlapping nodes were extracted from the FEM models of the horizontal structure samples to determine the interpass temperature before bead deposition and the cooling rates during bead deposition. The interpass temperatures for both samples are shown in Table 3 and Fig. 10. In addition, the data from the cooling curves were used to obtain the solidification rate from $T_{Liquidus}$ to $T_{Solidus}$ [43] which are 1450–1400 °C, respectively. The calculated cooling rates from the FEM model cooling curves for the continuous deposition horizontal structure, from bead 2–6, are 85.7 °C/s, 46 °C/s, 23.5 °C/s, 12.8 °C/s, and 7.9 °C/s, respectively. Due to high cooling rates in the intermittent deposition horizontal structure, the temperature–time data

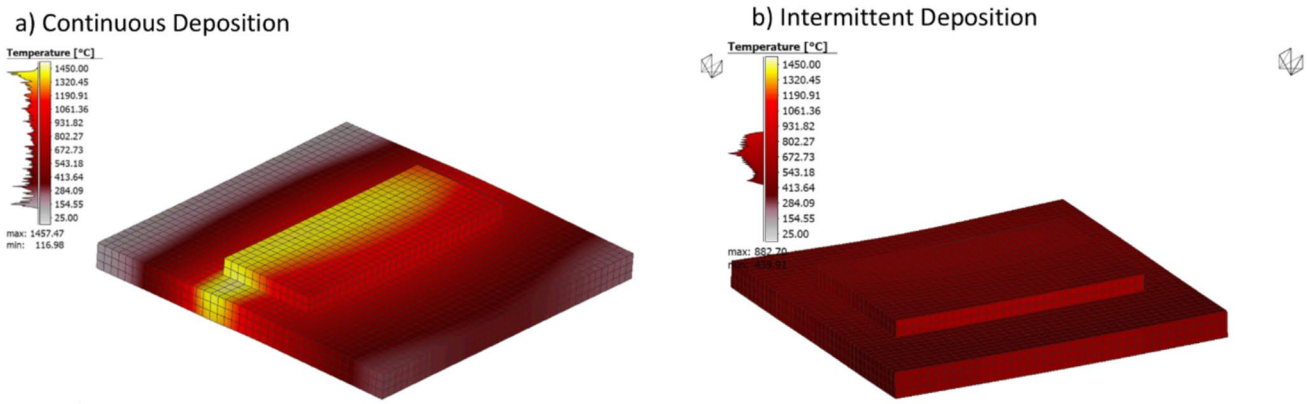


Fig. 9 Temperature distribution after the deposition of the last bead for **a** continuous deposition horizontal structure and **b** Intermittent-deposition horizontal structure

Table 3 Interpass temperatures for each bead for the investigated horizontal structure samples

Sample	Interpass Temperature (°C)					
	Bead 1	Bead 2	Bead 3	Bead 4	Bead 5	Bead 6
Continuous Deposition	25	664	986	1129	1259	1329
Intermittent Deposition	25	306	449	516	572	596

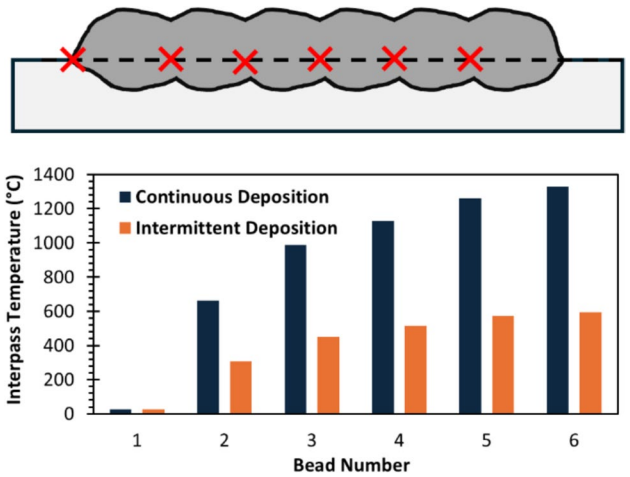


Fig. 10 Interpass temperature for continuous- and intermittent deposition horizontal structures

points acquired from the software were not sufficient to calculate the cooling rate. The interpass temperatures, shown in Fig. 10, indicate a higher interpass temperature for the second bead of the continuous deposition horizontal structure compared to the interpass temperatures of all the beads of the intermittent deposition horizontal structure as dwell time is the main factor in controlling the interpass temperature [44, 45]. Additionally, the continuous horizontal structure sample displays a significant increase in the interpass temperature compared to each consecutive layer due to the increase in the heat accumulation in the substrate [46] which is expected to affect the geometry of the deposited bead.

3.3 Macrostructure and microstructure evolution in WAAM horizontal structures

3.3.1 Macrostructure assessment

Figures 11 and 12 represent cross-sectional views of both horizontal samples continuously processed by WAAM and with the application of 60-s dwell time, respectively. It could be observed that the continuous processing of overlapping beads provides a significant penetration depth within the substrate, increasing the number of deposited beads. On the contrary, a dwell time of 60 s between the depositions of successive beads leads to a poor bonding with the substrate material in addition to a lack of fusion defects, as shown in the higher magnification micrographs of Fig. 12. This could be attributed to the relatively high cooling interval between the processing stages, and the heat input was not enough to melt the substrate for adequate fusion, especially for a high number of deposited beads.

Moreover, a statistical presentation of the height and depth of penetration of the deposited beads is given in Fig. 13. The intermittent deposition horizontal structure shows a decrease in the depth of penetration while an increase in the bead height with an approximate sum of the two lengths. With a constant heat input, any increase in bead height may correspond to a decrease in depth of penetration,

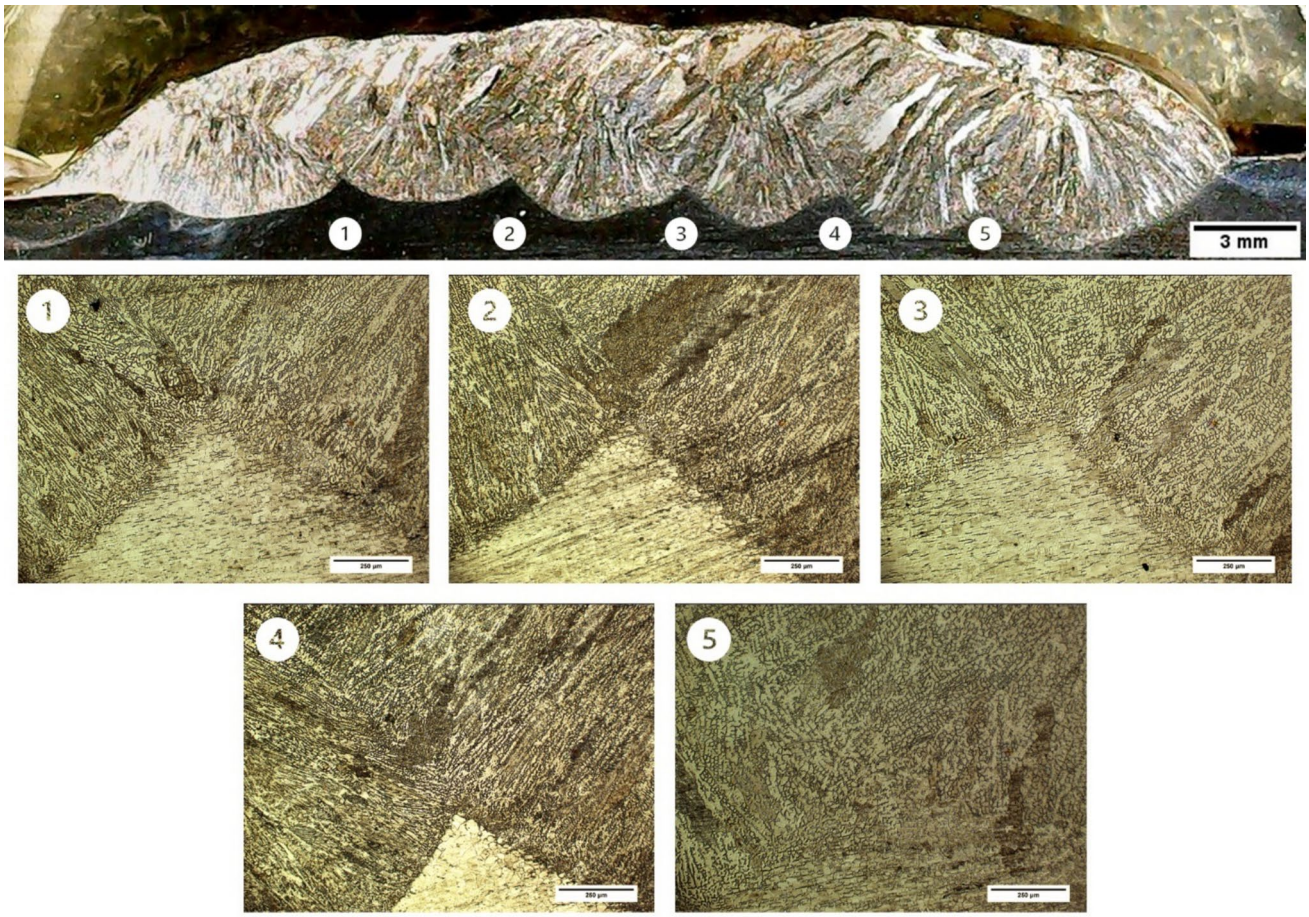


Fig. 11 Macro- and micrographs of continuous deposition horizontal structure (overlapping direction is from left to right)

and vice versa [47]. During overlap, the depth of penetration decreases because a part of the energy required to melt the substrate is employed to melt the overlapped portion of the previous clad, which will consequently increase the bead height [48]. On the contrary, the continuous horizontal structure sample shows an initial decrease in the penetration depth and an increase in the bead height in the second bead, followed by an evident increase in the penetration depth and slightly decreasing bead height. The first decrease in the penetration depth can be attributed to the low interpass temperature, which does not significantly affect the heat input. Furthermore, the interpass temperature drastically increases starting from bead three, leading to higher heat input and thus increasing the depth of penetration and reducing the bead height.

3.3.2 Microstructure assessment

Generally, epitaxial growth of columnar grains, which follows the heat gradient in the overlapping direction, is observed for both investigated conditions, Figs. 11 and 12. During the continuous processing of beads, it was found that the columnar structures are coarsened for each successive deposit due to the high heat accumulation and relatively lower cooling rates, while the application of dwell interval led to a finer structure. The polarized image of Fig. 14 highlights coarser columnar grains for the last deposit bead processed continuously compared to the initial bead. Moreover, the substructure morphologies and the dendritic spacing were found to vary drastically from fine lathy- and skeletal-delta ferrite in the initial bead to coarser skeletal delta ferrite in the last deposit, as shown in the higher magnification micrograph of Fig. 15.

For a quantitative assessment of the microstructure, the secondary dendritic arm spacing (SDAS) of the skeletal ferrite was measured at each overlapping node from the microstructure images of both investigated horizontal samples in Figs. 11 and 12, and the measurements are collected in Table 4. The initial SDAS in the 1st bead of both continuous deposition and intermittent deposition horizontal structures are comparable as the conditions during the deposition of the first bead are similar in both

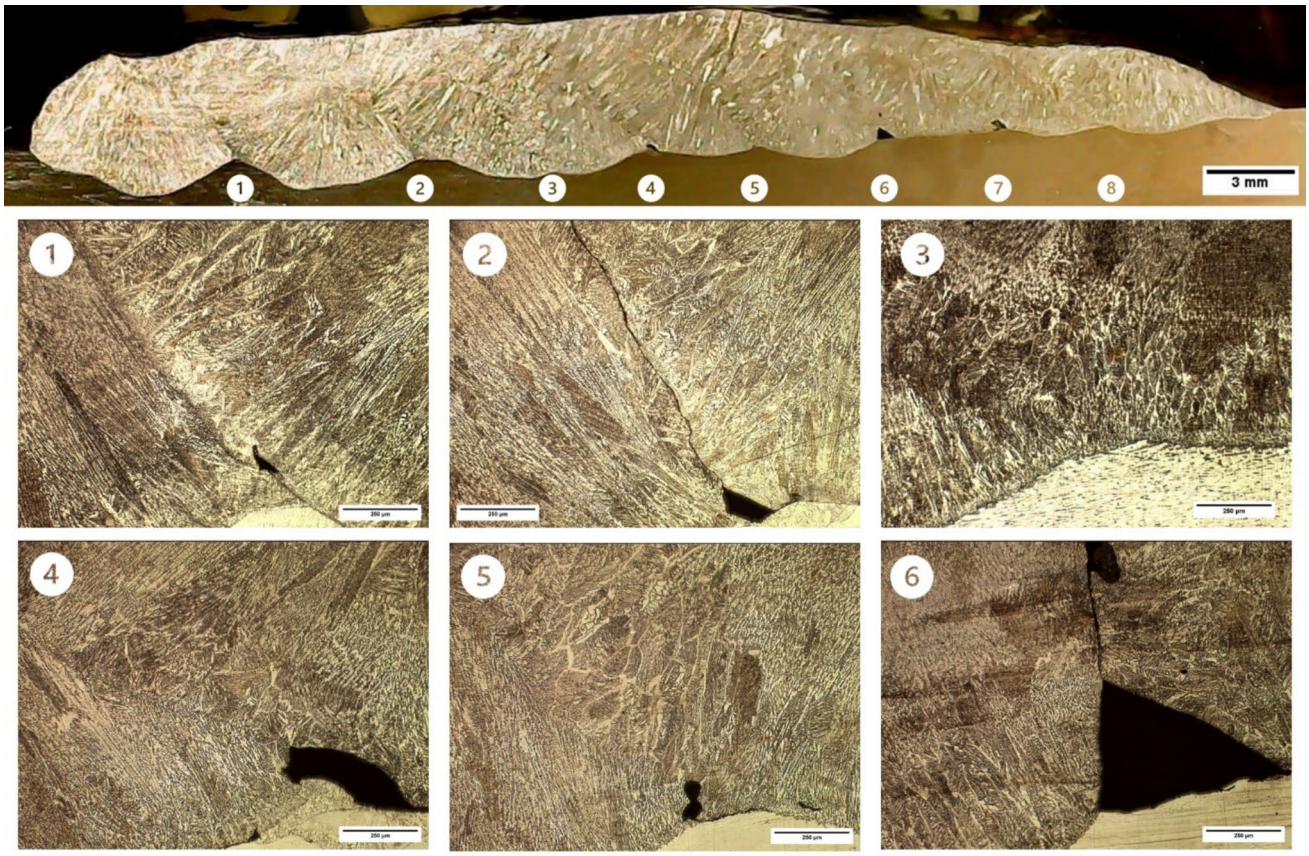


Fig. 12 Macro- and micrographs of intermittent deposition horizontal structure with 60 s dwell time (overlapping direction is from left to right)

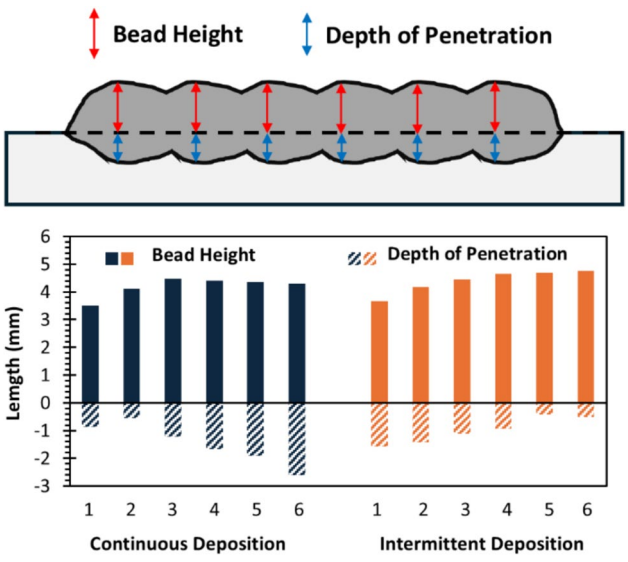


Fig. 13 Bead height and depth of penetration for WAAM horizontal structures processed by continuous deposition and intermittent deposition

structures. A significant increase in the measured SDAS is evident in the consecutive beads of the continuous deposition horizontal structure compared to the minor increase in the measured SDAS in the consecutive beads of the intermittent deposition horizontal structure. This difference can be attributed to the higher interpass temperatures recorded for the continuous deposition horizontal structure. Higher interpass temperatures result in lower cooling rates [45], which influences the development of the microstructure, leading to an increase of both primary and secondary dendrite arm spacings [46].

For SS308L, the relation between the secondary dendritic arm spacing (SDAS) and the cooling rate (CR) is [49]

$$\mathrm{SDAS} = 50\mathrm{CR}^{-0.4} \tag{4}$$

where SDAS is in μm and CR is in °C/s. The expected secondary dendritic arm spacing calculated from the cooling rates, using Eq. (4), extracted from the FEM model for the continuous horizontal structure sample was 8.4 μm, 10.8 μm, 13.4 μm, and 16.3 μm. The calculated values compared to the measured values from the FEM model are

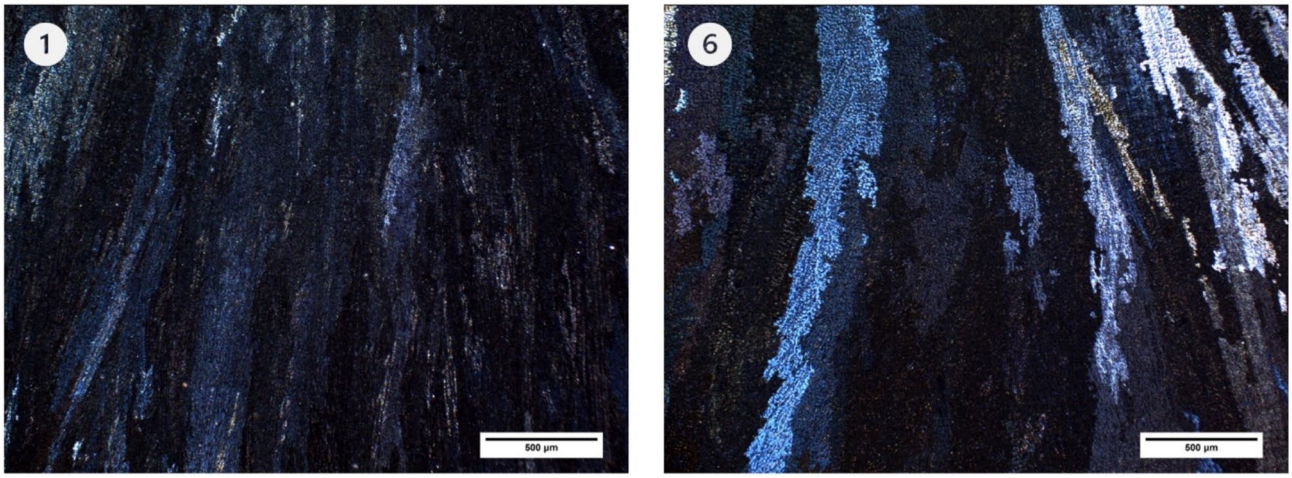


Fig. 14 Polarized light microscope images at the middle of the first and the sixth bead of the continuous deposition horizontal structure

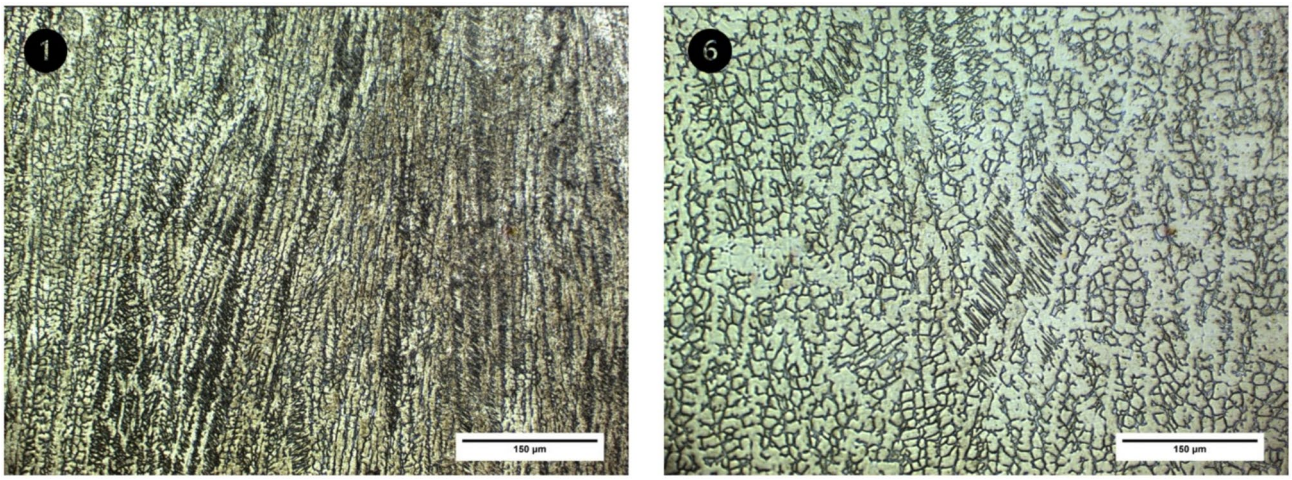


Fig. 15 Bright light microscope images at the middle of the first and the sixth bead of the continuous deposition horizontal structure

Table 4 Secondary dendritic arm spacing measurement for each bead for the investigated horizontal structures

Sample	Secondary Dendritic Arm Spacing (µm)				
	Bead 1	Bead 2	Bead 3	Bead 4	Bead 5
Continuous Deposition	5.4 ± 0.6	8.8 ± 1	11 ± 1	13.4 ± 0.9	16.2 ± 1.4
Intermittent Deposition	4.7 ± 0.9	6.4 ± 0.6	7.1 ± 0.4	8.2 ± 0.6	8.7 ± 0.4

approximately equal, as also shown in Fig. 16, validating the FEM model for cooling rate measurement. Furthermore, the cooling rates for the horizontal structure samples can thus be calculated from the SDAS-CR relation, shown in Table 5. Le et al.[38] have also observed a variation in the SDAS for the SS308L WAAM vertical structure of 18 layers processed with a dwell time of 60 s, in which the SDAS was found to be increasing upon successive deposition of layers (7.92 ± 0.45 µm and 11.24 ± 0.87 µm for the initial and last deposited track, respectively).

3.4 Hardness profile measurement

The SDAS significantly impacts mechanical properties, such as hardness and tensile strength. For instance, a decrease in SDAS leads to an increase in hardness due to the refinement of the dendritic structure, which reduces grain size and increases the resistance to deformation of the material [50]. Therefore, the hardness of each bead for the two horizontal structure samples was measured, as shown in Table 6 and Fig. 17. The measurements were

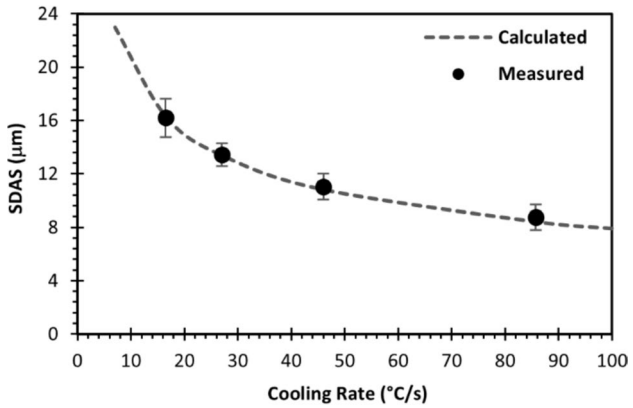


Fig. 16 Comparison between the relation of the calculated and measured SDAS against the cooling rate

Table 5 Cooling rate values obtained from the secondary dendritic arm spacing relation for each bead for the two horizontal structure samples

Sample	Cooling Rate (°C/s)				
	Bead 1	Bead 2	Bead 3	Bead 4	Bead 5
Continuous Deposition	261.7	78.0	43.9	26.8	16.8
Intermittent Deposition	375.3	170.2	130.5	90.5	79.3

taken in the middle of the bead above the substrate to measure for each bead separately without interfering with the consecutive bead and to avoid false readings from the higher hardness SS316L substrate.

Figure 17 shows that the hardness profile of the continuous deposition horizontal structure decreases with a higher slope than that of the intermittent deposition horizontal structure. Notably, the hardness of the sixth bead is higher than that of the fifth bead in the continuous horizontal structure sample. During WAAM, previously deposited layers undergo multiple thermal cycles that lead to a decrease in hardness. Thus, as the last deposited layer does not experience any repeated heating, it retains higher hardness than the previously deposited layer [51]. Furthermore, upon comparing the SDAS and the hardness of each bead, the hardness of each bead against the SDAS measured shows an expected inverse linear behavior, as shown in Fig. 18. Le et al. [38] have also found a declination in the hardness profile corresponding to the increase in SDAS for the studied SS308L vertical structure starting from approximately 168 HV0.1 to 151 HV0.1 from the bottom to the top, respectively.

4 Conclusion

The current work aims to thermally monitor the WAAM processing of SS308L vertical and horizontal structures through thermo-mechanical simulation models to manage heat accumulation and thus enhance product structure as well as properties with reduced experimental efforts. The models were validated by comparing the distortion in the substrate with the calculated distortion by the model. The main conclusions could be drawn as follows:

- The constructed FEM model offered comparable results to the thermal measurements, which could provide more reliable temperatures, especially at the deposition zones that are challenging to be captured by the mono-wavelength pyrometer due to the nature of arc generation during WAAM.
- By combining FEM predictions with mono-wavelength pyrometer readings, a master chart was developed, correlating the temperature of SS308L with emissivity variations where 600–1700 °C corresponds to 0.6–0.41, respectively. This approach addresses a significant limitation in in-process thermal monitoring using mono-wavelength pyrometers, enabling more accurate temperature estimations for thermal cycle measurement.
- The FEM model successfully predicted the significant heat accumulation resulting from the continuous deposition of horizontal structures, highlighting the necessity of dwell time for effective thermal management. Additionally, interpass temperatures and cooling rates calculated by the model provided accurate estimations of the resulting microstructure. The microstructure features had a direct correlation with mechanical properties, as reflected in the hardness profile measurements for both continuous and intermittent deposition conditions.
- From the initial to the last investigated bead, continuous deposition led to slower cooling rates, coarser columnar grains, a 200% increase in SDAS, and a 22.2% hardness reduction. In contrast, intermittent processing showed only an 85% SDAS increase and a

Table 6 Hardness measurement for each bead for the two horizontal structure samples

Sample	Hardness (HV0.5)					
	Bead 1	Bead 2	Bead 3	Bead 4	Bead 5	Bead 6
Continuous Deposition	189.4 ± 5.8	173.2 ± 3.9	163.8 ± 3.8	159.9 ± 5.6	147.4 ± 2.0	153.5 ± 2.9
Intermittent Deposition	185.5 ± 4.3	179.4 ± 5.8	173.0 ± 3.9	170.2 ± 5.6	168.4 ± 3.3	167.5 ± 3.7

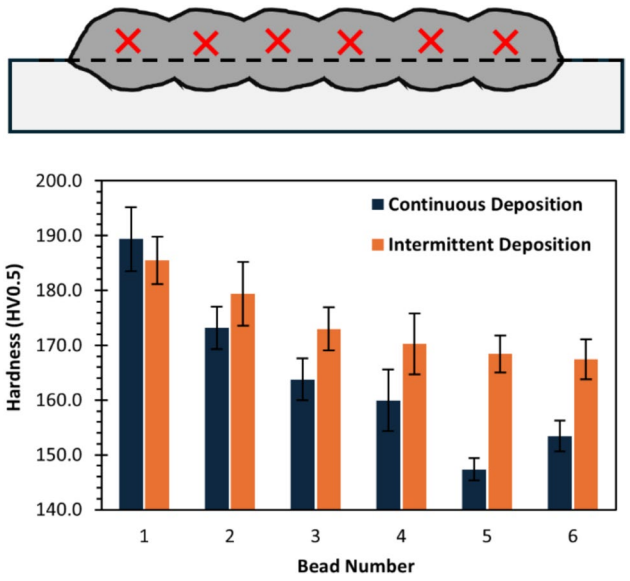


Fig. 17 Hardness profile for continuous deposition and intermittent deposition horizontal structure

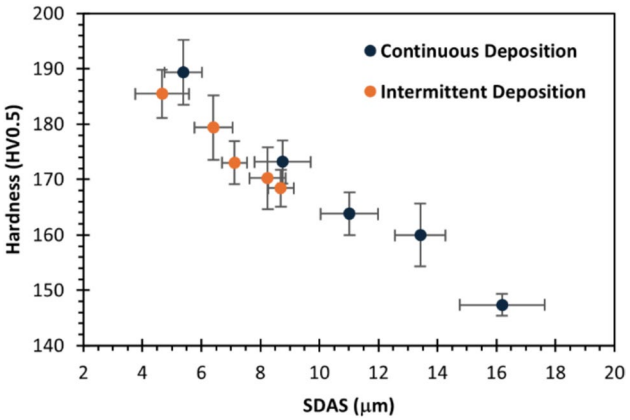


Fig. 18 The relation between bead hardness and secondary dendritic arm spacing

9.7% hardness reduction, highlighting the importance of thermal management in controlling properties during WAAM.

Acknowledgements The authors would like to acknowledge the Academy of Scientific Research and Technology, Egypt, for funding Prof. Mohamed A. Taha through the "Applied Research and Technology Short-Term Fellowship Program [ASRT-STARS]." The authors also extend their gratitude to Prof. J. Buhl at the Institute for Metallurgy, Materials Forming, Clausthal University of Technology, Germany, for the valuable discussions and facilities provided.

Author contributions Y.O. conducted the experiments, performed data analysis, developed the numerical simulations, and wrote the initial draft of the manuscript. M.A. conceptualized the research, contributed to manuscript writing, and reviewed the final draft. A.E. and M.A.T. provided supervision, critical review, and feedback on the manuscript.

Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB). This research was supported through the Short-Term Applied Research & Technology Transfer Fellowship Program [ASRT-STARS] provided by the Academy of Scientific Research & Technology.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Liu J, Xu Y, Ge Y, Hou Z, Chen S (2020) Wire and arc additive manufacturing of metal components: a review of recent research developments. Int J Adv Manuf Technol 111:149–198. https://doi.org/10.1007/s00170-020-05966-8
2. Lin Z, Song K, Yu X (2021) A review on wire and arc additive manufacturing of titanium alloy. J Manuf Process 70:24–45. https://doi.org/10.1016/j.jmapro.2021.08.018
3. Vimal KEK, Naveen Srinivas M, Rajak S (2021) Wire arc additive manufacturing of aluminium alloys: a review. Mater Today Proc 41:1139–1145. https://doi.org/10.1016/j.matpr.2020.09.153
4. Dhinakaran V, Ajith J, Fathima Yasin Fahmidha A, Jagadeesha T, Sathish T, Stalin B (2020) Wire arc additive manufacturing (WAAM) process of nickel based superalloys – a review. Mater Today Proc 21:920–925. https://doi.org/10.1016/j.matpr.2019.08.159
5. Ali Y, Henckell P, Hildebrand J, Reimann J, Bergmann JP, Barnikol-Oettler S (2019) Wire arc additive manufacturing of hot work tool steel with CMT process. J Mater Process Technol 269:109–116. https://doi.org/10.1016/j.jmatprotec.2019.01.034
6. Jin W, Zhang C, Jin S, Tian Y, Wellmann D, Liu W (2020) Wire arc additive manufacturing of stainless steels: a review. Appl Sci 10:1563. https://doi.org/10.3390/app10051563
7. Queguineur A, Cherukuri R, Lambai A, Dalal MS, Peura P, Mohanty G, Hascoët J-Y, Ituarte IF (2024) Correlated high throughput nanoindentation mapping and microstructural characterization of wire and arc additively manufactured 2205 duplex stainless steel. Weld World 68:2247–2257. https://doi.org/10.1007/s40194-024-01795-5
8. Bhadeshia H, Honeycombe R (2017) Chapter 12 - Stainless steel. In: Bhadeshia H, Honeycombe R (eds) Steels: microstructure and properties (Fourth Edition). Butterworth-Heinemann, pp 343–376

9. Lippold JC, Kotecki DJ (2005) Welding metallurgy and weldability of stainless steels. John Wiley & Sons
10. Ye C, Lu G, Peng X, Hou S, Zhou J, Ni L (2020) Microstructure and mechanical properties of the 316 stainless steel nuclear grade experimental component made by wire and arc additive manufacturing. Proc Inst Mech Eng C J Mech Eng Sci 234:4258–4267. https://doi.org/10.1177/0954406220920692
11. Gürol U, Kocaman E, Dilibal S, Koçak M (2023) A comparative study on the microstructure, mechanical properties, wear and corrosion behaviors of SS 316 austenitic stainless steels manufactured by casting and WAAM technologies. CIRP J Manuf Sci Technol 47:215–227. https://doi.org/10.1016/j.cirpj.2023.10.005
12. Taşdemir A, Nohut S (2021) An overview of wire arc additive manufacturing (WAAM) in shipbuilding industry. Ships Offshore Struct 16:797–814. https://doi.org/10.1080/17445302.2020.1786232
13. Pant H, Arora A, Gopakumar GS, Chadha U, Saeidi A, Patterson AE (2023) Applications of wire arc additive manufacturing (WAAM) for aerospace component manufacturing. Int J Adv Manuf Technol 127:4995–5011. https://doi.org/10.1007/s00170-023-11623-7
14. Josten A, Höfemann M (2020) Arc-welding based additive manufacturing for body reinforcement in automotive engineering. Weld World 64:1449–1458. https://doi.org/10.1007/s40194-020-00959-3
15. Evans SI, Wang J, Qin J, He Y, Shepherd P, Ding J (2022) A review of WAAM for steel construction – manufacturing, material and geometric properties, design, and future directions. Structures 44:1506–1522. https://doi.org/10.1016/j.istruc.2022.08.084
16. Cunningham CR, Flynn JM, Shokrani A, Dhokia V, Newman ST (2018) Invited review article: Strategies and processes for high quality wire arc additive manufacturing. Addit Manuf 22:672–686. https://doi.org/10.1016/j.addma.2018.06.020
17. Ozaner OC, Klobčar D, Sharma A (2023) Machining strategy determination for single- and multi-material wire and arc additive manufactured thin-walled parts. Materials 16:2055. https://doi.org/10.3390/ma16052055
18. Shah A, Aliyev R, Zeidler H, Krinke S (2023) A review of the recent developments and challenges in wire arc additive manufacturing (WAAM) process. J Manuf Mater Process 7:97. https://doi.org/10.3390/jmmp7030097
19. Nagallapati V, Khare VK, Sharma A, Simhambhatla S (2023) Active and passive thermal management in wire arc additive manufacturing. Metals (Basel) 13:682. https://doi.org/10.3390/met13040682
20. Wang S, Gu H, Wang W, Li C, Ren L, Wang Z, Zhai Y, Ma P (2020) The influence of heat input on the microstructure and properties of wire-arc-additive-manufactured Al-Cu-Sn alloy deposits. Metals (Basel) 10:79. https://doi.org/10.3390/met10010079
21. Li C, Gu H, Wang W, Wang S, Ren L, Wang Z, Ming Z, Zhai Y (2019) Effect of heat input on formability, microstructure, and properties of Al–7Si–0.6Mg alloys deposited by CMT-WAAM process. Appl Sci 10:70. https://doi.org/10.3390/app10010070
22. Rosli NA, Alkahari MR, bin Abdollah MF, Maidin S, Ramli FR, Herawan SG (2021) Review on effect of heat input for wire arc additive manufacturing process. J Mater Res Technol 11:2127–2145. https://doi.org/10.1016/j.jmrt.2021.02.002
23. Chen F, Yang Y, Feng H (2021) Regional control and optimization of heat input during CMT by wire arc additive manufacturing: modeling and microstructure effects. Materials 14:1061. https://doi.org/10.3390/ma14051061
24. Kozamernik N, Bračun D, Klobčar D (2020) WAAM system with interpass temperature control and forced cooling for near-net-shape printing of small metal components. Int J Adv Manuf Technol 110:1955–1968. https://doi.org/10.1007/s00170-020-05958-8
25. Azizul Izham EDI, Alkahari MR, Hussein NIS, Maidin S, Ramli FR, Herawan SG (2023) Online monitoring of wire arc additive manufacturing process: a review. Adv Mater Process Technol. https://doi.org/10.1080/2374068X.2023.2190669
26. Ck N, Jhavar S (2025) Substrate-centric comparative analysis and validation of finite element and empirical models for residual stress prediction in remanufactured WAAM-fabricated SS316 parts. Multiscale Multidiscip Model Exp Des 8:112. https://doi.org/10.1007/s41939-024-00690-2
27. Béraud N, Ledoux Y, Mechekour EH, Vignat F, Pourroy F (2024) Fast thermal simulation of WAAM processing: toward manufacturing strategy evaluation. CIRP J Manuf Sci Technol 55:234–246. https://doi.org/10.1016/j.cirpj.2024.10.007
28. Scotti FM, Teixeira FR, da Silva LJ, de Araújo DB, Reis RP, Scotti A (2020) Thermal management in WAAM through the CMT advanced process and an active cooling technique. J Manuf Process 57:23–35. https://doi.org/10.1016/j.jmapro.2020.06.007
29. Jorge VL, Teixeira FR, Scotti A (2022) Pyrometrical interlayer temperature measurement in WAAM of thin wall: strategies, limitations and functionality. Metals (Basel) 12:765. https://doi.org/10.3390/met12050765
30. Pixner F, Buzolin R, Schönfelder S, Theuermann D, Warchomicka F, Enzinger N (2021) Contactless temperature measurement in wire-based electron beam additive manufacturing Ti-6Al-4V. Weld World 65:1307–1322. https://doi.org/10.1007/s40194-021-01097-0
31. Halisch C, Radel T, Tyralla D, Seefeld T (2021) Online monitoring in WAAM. PhotonicsViews 18:2–3. https://doi.org/10.1002/phvs.202100005
32. Müller J, Hensel J (2024) Potential of thermography for the monitoring of DED-arc processes. Weld World 68:505–513. https://doi.org/10.1007/s40194-023-01676-3
33. Huang H, Ma N, Chen J, Feng Z, Murakawa H (2020) Toward large-scale simulation of residual stress and distortion in wire and arc additive manufacturing. Addit Manuf 34:101248. https://doi.org/10.1016/j.addma.2020.101248
34. Li R, Xiong J (2019) Influence of interlayer dwell time on stress field of thin-walled components in WAAM via numerical simulation and experimental tests. Rapid Prototyp J 25:1433–1441. https://doi.org/10.1108/RPJ-03-2019-0067
35. Yang D, Xiong J, Li R (2020) Numerical study of heat transfer characteristics in positional wire and arc additive manufacturing. Rapid Prototyp J 26:1627–1635. https://doi.org/10.1108/RPJ-01-2020-0016
36. Abusalma H, Eisazadeh H, Hejripour F, Bunn J, Aidun DK (2022) Parametric study of residual stress formation in wire and arc additive manufacturing. J Manuf Process 75:863–876. https://doi.org/10.1016/j.jmapro.2022.01.043
37. Kumar V, Anand M, Mandal A (2024) Critical analysis of interlayer dwell time on the properties of wire arc additive manufactured steel structure: experimental and modelling approach. Arab J Sci Eng. https://doi.org/10.1007/s13369-024-09120-4
38. Le VT, Mai DS, Paris H (2021) Influences of the compressed dry air-based active cooling on external and internal qualities of wire-arc additive manufactured thin-walled SS308L components. J Manuf Process 62:18–27. https://doi.org/10.1016/j.jmapro.2020.11.046
39. Heuser BJ, Li Z, Bai X-M, Zhong W, Sutton BJ (2023) Irradiation-assisted cracking of SA508-304L weldments with 308L groove filler and 309L butter in hot water immersion constant rate extension tests. J Nucl Mater 581:154472. https://doi.org/10.1016/j.jnucmat.2023.154472
40. Bernhard F (2014) Handbuch der Technischen Temperaturmessung. Springer, Berlin Heidelberg, Berlin, Heidelberg
41. Nomura K, Okuda H, Sano T, Asai S (2022) Simultaneous multipoint emissivity measurement via Zebra-patterned blackbody

spray method and application to gas tungsten arc welding process. J Manuf Process 78:22–31. https://doi.org/10.1016/j.jmapro.2022.04.004
42. Felice RA, Nash DA (2013) Pyrometry of materials with changing, spectrally-dependent emissivity-solid and liquid metals. AIP Conf Proc 1552(8):734–739. https://doi.org/10.1063/1.4819633
43. Jeong J, Webster S, Liao S, Mogonye JE, Ehmann K, Cao J (2022) Cooling rate measurement in directed energy deposition using photodiode-based planck thermometry (PDPT). Addit Manuf Lett. https://doi.org/10.1016/j.addlet.2022.100101
44. Zhai W, Wu N, Zhou W (2022) Effect of interpass temperature on wire arc additive manufacturing using high-strength metal-cored wire. Metals (Basel) 12:212. https://doi.org/10.3390/met12020212
45. Wu B, Ding D, Pan Z, Cuiuri D, Li H, Han J, Fei Z (2017) Effects of heat accumulation on the arc characteristics and metal transfer behavior in wire arc additive manufacturing of Ti6Al4V. J Mater Process Technol 250:304–312. https://doi.org/10.1016/j.jmatprotec.2017.07.037
46. Zhang C, Qiu Z, Zhu H, Wang Z, Muránsky O, Ionescu M, Pan Z, Xi J, Li H (2021) On the effect of heat input and interpass temperature on the performance of inconel 625 alloy deposited using wire arc additive manufacturing-cold metal transfer process. Metals (Basel) 12:46. https://doi.org/10.3390/met12010046
47. Sugitani Y, Kanjo Y, Nishi Y (1989) Simultaneous control of penetration depth and bead height by controlling multiple welding parameters. Q J Jpn Weld Soc 7:21–26. https://doi.org/10.2207/qjjws.7.21
48. Ramiro P, Ortiz M, Alberdi A, Lamikiz A (2020) Strategy development for the manufacturing of multilayered structures of variable thickness of Ni-based alloy 718 by powder-fed directed energy deposition. Metals (Basel) 10:1–18. https://doi.org/10.3390/met10101280
49. Le VT, Mai DS, Bui MC, Wasmer K, Nguyen VA, Dinh DM, Nguyen VC, Vu D (2022) Influences of the process parameter and thermal cycles on the quality of 308L stainless steel walls produced by additive manufacturing utilizing an arc welding source. Weld World 66:1565–1580. https://doi.org/10.1007/s40194-022-01330-4
50. Tandon V, Thombre MA, Patil AP, Taiwade RV, Vashishtha H (2020) Effect of heat input on the microstructural, mechanical, and corrosion properties of dissimilar weldment of conventional austenitic stainless steel and low-nickel stainless steel. Metallogr Microstruct Anal 9:668–677. https://doi.org/10.1007/s13632-020-00681-y
51. Dekis M, Tawfik M, Egiza M, Dewidar M (2024) Unveiling the characteristics of ER70S-6 low carbon steel alloy produced by wire arc additive manufacturing at different travel speeds. Met Mater Int. https://doi.org/10.1007/s12540-024-01766-x

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.